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Class : AIDS

Sub : Python for Data Science

Aim: 6.Implementing Web Scraping in Python with BeautifulSoup.

Theory:

1. What is Web Scraping? Role of Requests Library

Web Scraping: Web scraping is the process of extracting data from websites automatically. It helps collect information like text, images, prices, etc. from web pages.

Role of Requests Library: The Requests library in Python is used to send HTTP requests to websites. It helps to: Fetch webpage content (HTML) Send GET/POST requests Handle headers and cookies Example: `requests.get(url)` is used to download a webpage.

2. What is BeautifulSoup? How it helps in parsing HTML?

BeautifulSoup: BeautifulSoup is a Python library used for parsing HTML and XML documents. How it helps: Converts raw HTML into a structured format (parse tree)

Makes it easy to: Navigate HTML elements Search for specific tags Extract required data

3. Difference between Web Scraping and Web Crawling

Web Scraping: Used to extract specific data from websites Focuses on collecting required information Works on particular web pages Used for data analysis, price tracking, etc. Web Crawling: Used to browse and discover web pages Focuses on indexing and scanning websites Automatically visits multiple pages using links Used by search engines like Google

4. Commonly Used Methods in BeautifulSoup

`find()` → Finds first matching tag
`find_all()` → Finds all matching tags
`get_text()` → Extracts text from HTML
`get()` → Gets attribute value (like href)
`select()` → Uses CSS selectors
`prettify()` → Formats HTML structure

5. Ethical and Legal Considerations in Web Scraping

Always check the website's robots.txt Do not overload servers (avoid too many requests) Respect copyright and data ownership Avoid scraping personal or sensitive data Follow website terms and conditions Use scraped data responsibly

```

import requests from bs4 import BeautifulSoup

# Step 1: Send request to the website
url = "https://example.com"
response = requests.get(url)

# Step 2: Parse the HTML content
soup = BeautifulSoup(response.content, 'html.parser')

# Step 3: Extract data # Example 1: Get page title
title = soup.title.string
print("Page Title:", title)

# Example 2: Get all links
links = soup.find_all('a')
print("\nAll Links:")
for link in links:
    print(link.get('href'))

# Example 3: Get all paragraphs
paragraphs = soup.find_all('p')
print("\nParagraphs:")
for p in paragraphs:
    print(p.get_text())

```

...Page title: Example Domain

All Links:

<https://iana.org/domains/example>

Paragraphs:

This domain is for use in documentation examples without needing permission.

Avoid use in operations.

[Learn more](#)

Conclusion:

Web scraping is a powerful technique for extracting data from websites. The Requests library is used to fetch webpage content, while BeautifulSoup helps in parsing and extracting data easily. It is widely used in data analysis, research, and automation tasks. However, scraping should always be done ethically by following website rules and legal guidelines.

